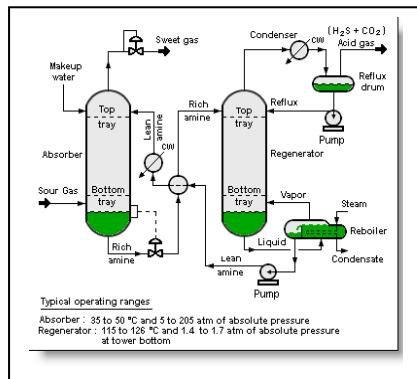




College:
Department:
Course:
Classroom and Time:

Instructor:
Office hours:
E-mail:
Cell Phone:

TA:
E-mail:
Office hours:

UIC COLLEGE OF ENGINEERING
UIC Department of Chemical Engineering
CHE 433: Process Simulation with ASPEN Plus
EIB 124: M/W: 03:30 PM – 04:45 PM

Dr. Michael Caracotsios, EIB 244
M|Tu|W|Th: Stop by my office or schedule a ZOOM meeting
mcaracot@uic.edu
(630) 696-0934

COURSE OBJECTIVES

This course presents the main principles of Process Simulation (Material and Energy Balances) relevant to chemical and biological processes. Process Simulation or Flowsheeting is usually defined as the use of commercially available products (like ASPEN Plus, ASPEN Hysys, UniSim, VMG, ChemCad, gProms, Pro II, etc.) to perform steady-state Heat and Weight Balance as well Sizing and Cost calculations for chemical, biological, petrochemical and refining processes. By the end of the semester the students should be able to:

- ✿ Select and justify choices for physical property models required for accurate heat and weight balance calculations.
- ✿ Design and model unit operations such as chemical reactors, heat exchangers, heaters and coolers, pumps, compressors and turbines, mixers, distillation and packed columns and user specific operations.
- ✿ Combine unit operations in a cohesive Flowsheet that describes existing or conceptual processes.
- ✿ Familiarize themselves with computational aspects such as design specs, calculator blocks, sensitivity analyses, flow sheet convergence
- ✿ Perform optimization studies.

COURSE PREREQUISITES

The following are key requirements for a successful completion of this course.

- **Mathematics:** Differential and Integral Calculus
- **Statistics:** Linear and Nonlinear Regression
- Numerical Methods for Chemical Engineering
- **Programming Experience:** Excel VBA, MATLAB, C++
- **CHE Department Prerequisites:** CHE 301, CHE 311, CHE 312, CHE 313 and CHE 321

SOFTWARE

All students must download and install **ASPEN Plus v14.0** and the Microsoft EXCEL **[32-bit Version] CHE Calculator v12.0** following the instructions listed on the UIC Blackboard in the **Computational Tools** content area of the course.

TEXTBOOK

Michael Caracotsios, *Process Simulation with ASPEN Plus*, 1st Edition, 2025 will be distributed to all students free of charge via the UIC Blackboard **Course Textbook** content area.

ADDITIONAL REFERENCES

1. Thomas A. Adams II, *Learn ASPEN PLUS in 24 Hours*, McGraw-Hill Education. (2018)
2. Ralph Schefflan, *Teach yourself the Basics of Aspen Plus*, 2nd Edition, Wiley. (2016).
3. Gavin Towler and Ray Sinnott, *Chemical Engineering Design*, 2nd Edition, Elsevier Ltd. (2013).
4. Lorenz T. Biegler, Ignacio E. Grossmann and Arthur W. Westerberg, *Systematic Methods of Chemical Process Design*, Prentice Hall International Series, December (1999).

CHE 433: COURSE OUTLINE

1. CHAPTER 1: REVIEW

- Mass, Energy, Entropy and Lost Work Balance Equations
- Calculus and Numerical Methods for Chemical Engineers

2. CHAPTER 2: INTRODUCTION TO PROCESS FLOWSHEETING

- Principles of Process Flowsheeting in Chemical Engineering
- First look at Tear Equations and Design Specifications
- Methods for Convergence and Optimization

3. CHAPTER 3: BASIC PROCESS UNIT OPERATIONS

- Getting Started with the **ASPEN Plus v14.0** Process Simulator
- Mixers and Splitters
- Separators: Two and Three Phase Flash Blocks
- Exchangers: Heaters, Coolers and Condensers
- Pressure Changers: Valves, Pumps and Compressors and Turbines

4. CHAPTER 4: AUXILIARY OPERATIONS & MODEL ANALYSIS TOOLS

- Getting Started with the **ASPEN Plus v14.0** Process Simulator
- Flowsheeting Options: Design Specs, Calculator and Transfer Blocks
- Model Analysis Tools: Sensitivity Analysis, Optimization and Data Fit
- Manipulators: Mult, Dupl, ClChng
- User Models: User2, Hierarchy

5. CHAPTER 5: PROCESS FLOWSHEETING FUNDAMENTALS

- Component and Stream Classes
- Selection of Physical Property Methods
- Recycle Stream Selection and Tear Equations

6. CHAPTER 6: DISTILLATION AND ABSORPTION

- Shortcut and Rigorous Distillation
- Absorption and Stripping Operations
- Homogeneous and Heterogeneous Azeotropic Distillation

7. CHAPTER 7: SIMULATION AND DESIGN OF HEAT EXCHANGERS

- Double-Pipe Heat Exchangers
- Shell and Tube Heat Exchanger Design
- Multi-Stream Heat Exchangers in Cryogenic Operations

8. CHAPTER 8: CHEMICAL KINETICS AND REACTOR UNIT OPERATIONS

- Yield, Conversion and Equilibrium Reactors
- CSTR, Plug Flow and Fixed Bed Reactors

9. CHAPTER 9: SPECIAL TOPICS IN PROCESS SIMULATION

- Solids Handling Operations and Electrolyte Systems
- Characterization of Drug Like Molecules
- ASPEN Process Economic Analyzer: Capital and Operating Costs Calculations

CHE 433: COURSE GRADING

Your final grade for the **CHE 433** Process Simulation with ASPEN Plus course will be determined from ten (10) take home assignments, quiz/attendance records and the final project. The contribution of all these activities to the grade calculation formula is shown below:

1. Home Assignments: [40%]
2. Final Project: [40%]
3. Quizzes & Attendance: [20%]

The formula for the calculation of your total score T is given by the equation below. The grades associated to a total score range are also given in the table below.

$$T(\text{Total Score}) = 0.40\langle H \rangle + 0.20\langle Q + A \rangle + 0.40\langle P \rangle$$

Total Score	Grade	Symbol	Max Score	Meaning of Symbols
$90 \leq T \leq 100$	A	$\langle H \rangle$	100 pts	Average Score for Assignments
$80 \leq T < 90$	B	$\langle Q + A \rangle$	100 pts	Average Score for Attendance
$70 \leq T < 80$	C	$\langle P \rangle$	100 pts	Score for the Final Project

Note 1: Completion of all Assignments is mandatory. In the event that a student does not complete an Assignment by the designated time and date, the student will receive a score of -100 points. The student may still be able to negotiate a better score by turning in the missed assignment at a time agreed upon with me. If attendance is taken and a student is absent the student will receive a score of -100 points, unless there is a reasonable explanation for the absence.

Note 2: Extra credit may be given at my discretion. The extra credit will be based on the student's record of attendance, punctuality, participation in class, good work and ethics behavior, and reasonable scores on all assignments and examinations. The extra credit will be contingent on perfect attendance and completion of all home assignments.

Dr. Michael Caracotsios
CLINICAL ASSOCIATE PROFESSOR
CHEMICAL ENGINEERING DEPARTMENT
UNIVERSITY OF ILLINOIS CHICAGO

PHONE: +1 (630) 696-0934
Email: mcaracot@uic.edu

CHE 433: MISCELLANEOUS

Miscellaneous information about this course, the policy on partial credit as well as the policies on student courtesy, tardiness and absenteeism is given in the sections below. Contact Professor Michael Caracotsios if you have any questions.

BLACKBOARD COLLABORATE HELP

By email: LTS@uic.edu

By phone: (312) 413-0003 Option #1

ACADEMIC INTEGRITY

As an academic community, UIC is committed to providing an environment in which research, learning, and scholarship can flourish and in which all endeavors are guided by academic and professional integrity. All members of the campus community—students, staff, faculty, and administrators—share the responsibility of insuring that these standards are upheld so that such an environment exists. Instances of academic misconduct by students will be handled pursuant to the Student Disciplinary Policy: <https://dos.uic.edu/conductforstudents.shtml>

RELIGIOUS HOLIDAYS

Students who wish to observe their religious holidays shall notify the faculty member by the tenth day of the semester of the date when they will be absent unless the religious holiday is observed on or before the tenth day of the semester. In such cases, the student shall notify the faculty member at least five days in advance of the date when he/she will be absent. The faculty member shall make every reasonable effort to honor the request, not penalize the student for missing the class, and if an examination or project is due during the absence, give the student an exam or assignment equivalent to the one completed by those students in attendance. If the student feels aggrieved, he/she may request remedy through the campus grievance procedure. The University Holidays and the Religious Observances calendar can be found online at the following website: <http://oae.uic.edu/religious-calendar/>. The list includes national holidays recognized by the University as well as religious days of special observance that may prohibit a student from attending classes. Please keep in mind this list is not exhaustive.

PARTIAL CREDIT POLICY

Partial credit can be requested under certain circumstances. The student should make an appointment with the Professor and be prepared to discuss the reasons for requesting the extra credit. Final decision rests with the Professor.

No partial credit will be given for:

1. Incorrect formulas, full or partial.
2. Inconsistent or missing units in calculations.
3. Multiple answers.
4. Incoherent (illogical, disjointed) derivations or calculations.

STUDENT COURTESY POLICY

Talking/Smart Phones/Laptops/Food and Drinks in Class/Tardiness:

Not allowed unless explicitly stated

Repeat Offenders will be asked to leave the classroom

Once instruction begins the class doors will close

Cheating/Copying:

Perpetrators will receive zero points in their homework

Homework contribution to the final grade is substantially reduced after first warning

Late Home Assignments and Disputes:

Late Home Assignments will not be accepted without prior arrangements with the Professor.

If there are Home Assignments score disputes, they must be resolved first with the TA.

If there are Exam score disputes, they will be resolved only with the Professor.